Chapter 5: The multiple regression model

Financial Econometrics

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5.1 Introduction

A general multiple regression model

A dependent variable y is related to a number of explanatory variables $x_2, \ldots, x_K$ through a linear equation that can be written as

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + e_i,$$

where $e_i \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2)$.

The multiple regression function (or simply the regression function) is

$$E(y \mid x_2, \ldots, x_K) = \beta_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_K x_K\,.$$

A single parameter, call it β_k, measures the effect of a change in the variable x_k upon the expected value of y, with all other variables held constant

$$\beta_k = \frac{\partial E(y \mid x_1, \ldots, x_K)}{\partial x_k}$$

Assumptions

MR1: Econometric model observations on $(y_i, x_i) = (y_i, x_{i2}, x_{i3}, \ldots, x_{iK})$ satisfy the population relationship,

$$y_i = \beta_1 + \beta_2 x_{i2} + \beta_3 x_{i3} + \cdots + \beta_K x_{iK} + e_i \,.$$

MR2: Strict Exogeneity: $E(e_i \,|\, X) = 0$.

MR3: Conditional Homoskedasiticity: $Var(e_i \,|\, X) = \sigma^2$.

MR4: Conditional Uncorrelated Errors: $Cov(e_i, e_j \,|\, X) = 0$ for $i \neq j$.

MR5: No Exact Linear Relationship Exists Between the Explanatory Variables

MR6: Error Normality (optional) Conditional on X : $e_i \,|\, X \sim N(0,\sigma^2)$.

Example

In the andy data, we consider a multiple linear regression model:

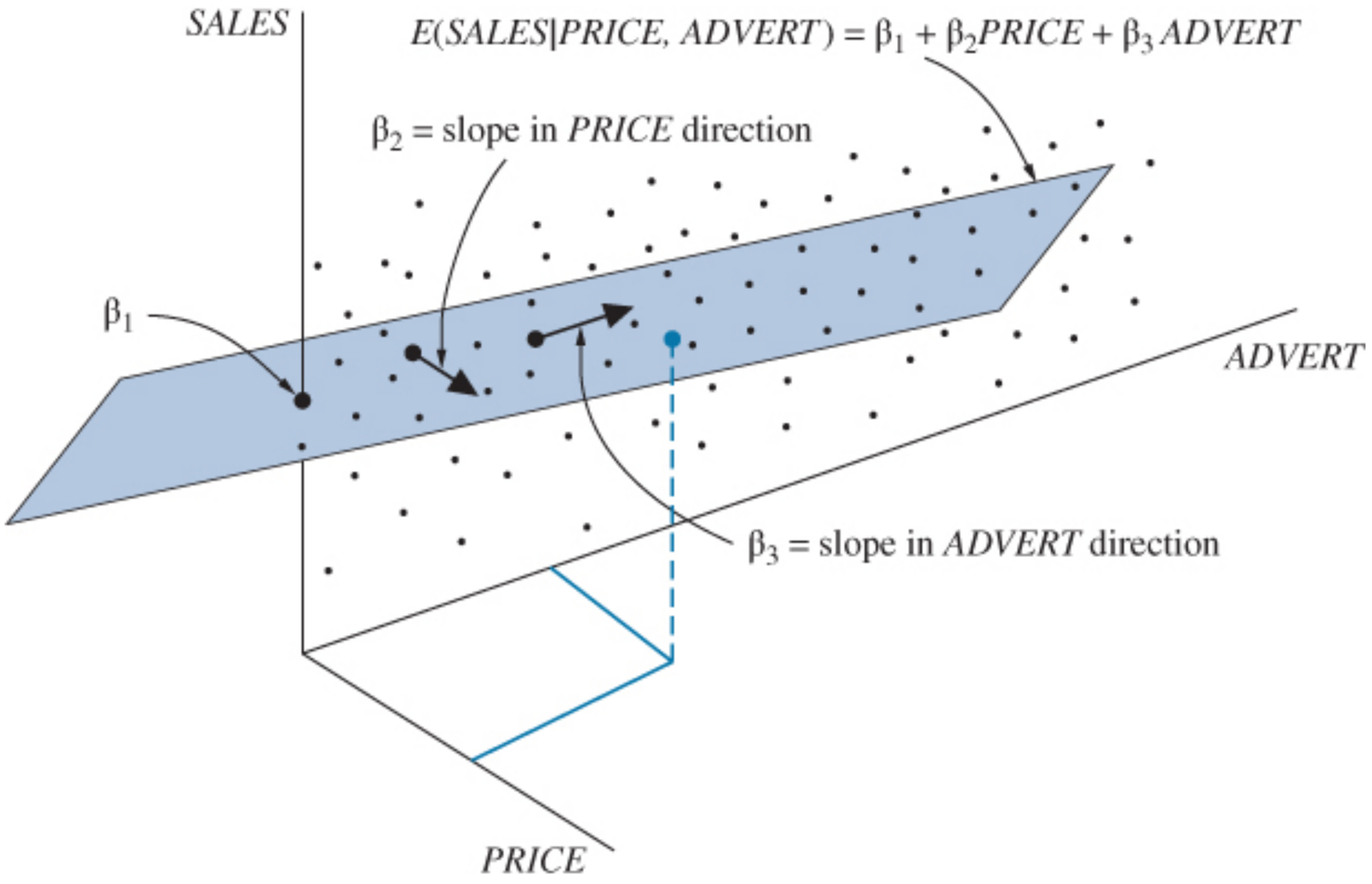
$$SALES = \beta_1 + \beta_2 PRICE + \beta_3 ADVERT + e\,.$$

The regression function is

$$E(SALES \mid PRICE, ADVERT) = \beta_1 + \beta_2 PRICE + \beta_3 ADVERT\,.$$

β_2 is the change in month sales $SALES$(\$1000) when the price index is increased by one unit (\$1), and advertising expenditure (ADVERT) is held constant.

β_3 is the change in month sales $SALES$(\$1000), when the advertising expenditure is increased by one unit(\$1000), and the price index is held constant.


SALES
E(SALES|PRICE, ADVERT) = β1 + β2PRICE + β3 ADVERT
β2 = slope in PRICE direction
β1
ADVERT
β3 = slope in ADVERT direction
PRICE

5.2 Estimating the parameters of the multiple regression model

Least squares estimation i

The least squares estimators $b_1, \ldots, b_k$ minimizes

$$S(\beta_1, \ldots, \beta_k) = \sum_{i=1}^{N} (y_i - E(y_i))^2$$
$$= \sum_{i=1}^{N} (y_i - \beta_1 - \beta_2 x_{i2} - \ldots - \beta_K x_{iK})^2 \,.$$

We estimate the error variance σ^2 by

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{N} \hat{e}_i^2}{N - K},$$

where the residual is

$$\hat{e}_i = y_i - \hat{y}_i = y_i - (b_1 + b_2 x_{i2} + \cdots + b_k x_{iK}) \,.$$

Example: `andy`

For any given (but fixed) value of advertising expenditure, an increase in price by $1 decreases expected sales by approximately $7908.

On the other hand, for any given price, expected sales increase approximately by $1863 when advertising expenditures increase by $1000.

R scripts i

```
> data(andy)
> tidy(andy) # summary statitsics for the data. need 'broom' package
> s=tidy(andy)[,c(1:5,8,9)]
> kable(s,caption="Summary statistics for dataset $andy$")
|column |  n|     mean|        sd| median|   min|   max|
|:------|--:|--------:|---------:|------:|-----:|-----:|
|sales  | 75| 77.37467| 6.4885365|  76.50| 62.40| 91.20|
|price  | 75|  5.68720| 0.5184320|   5.69|  4.83|  6.49|
|advert | 75|  1.84400| 0.8316769|   1.80|  0.50|  3.10|
>
> mod1 <- lm(sales~price+advert, data=andy)
> smod1 <- data.frame(xtable(summary(mod1)))
> kable(smod1,
+ caption="The basic multiple regression model",
```

R scripts i i

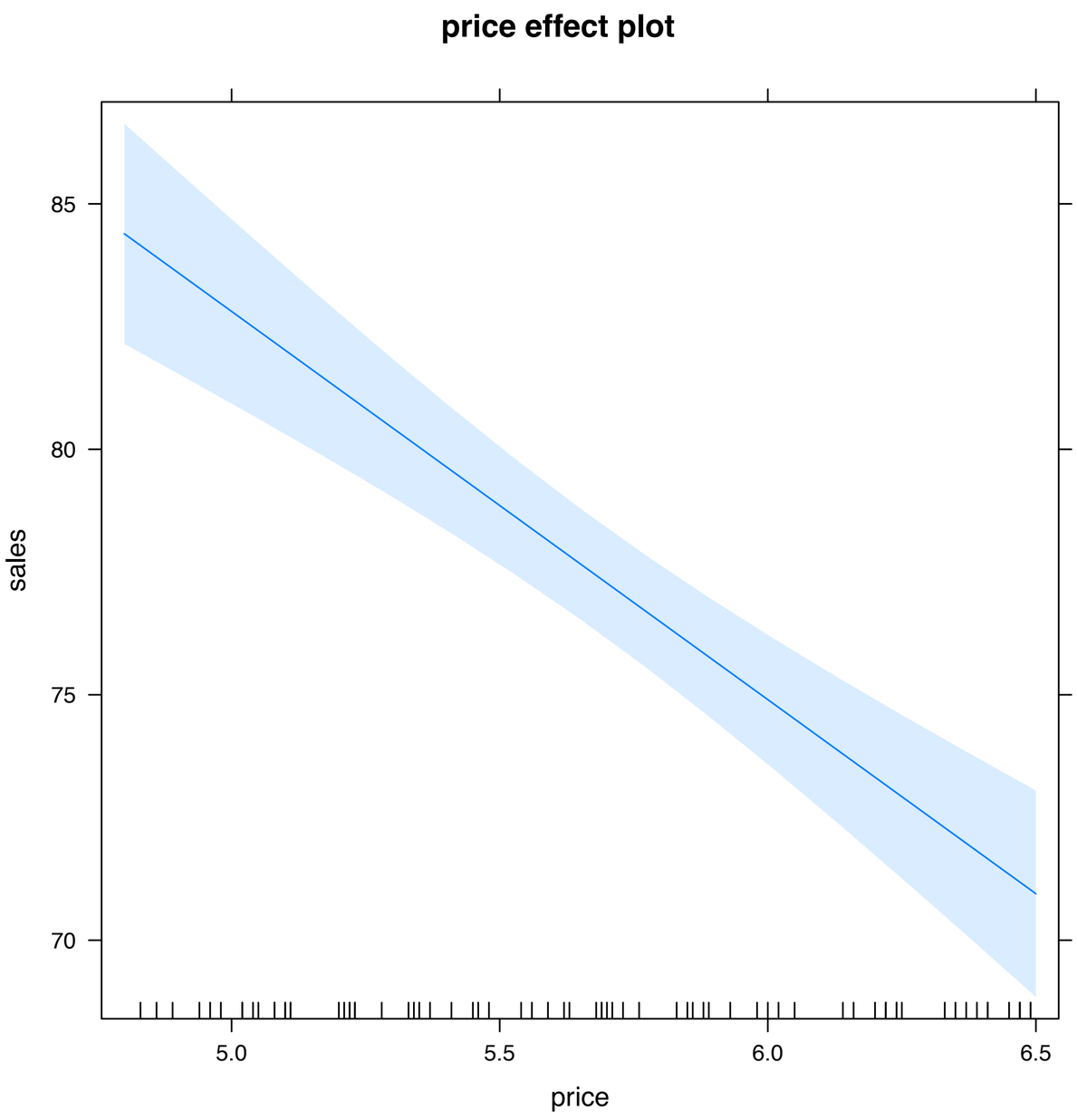
```
+ caption="The basic multiple regression model",
+ col.names=c("coefficient", "Std. Error", "t-value", "p-value"),
+ align="c", digits=3)
|            | coefficient | Std. Error | t-value | p-value |
|:-----------|:-----------:|:----------:|:-------:|:-------:|
|(Intercept) |   118.914   |   6.352    | 18.722  |  0.000  |
|price       |   -7.908    |   1.096    | -7.215  |  0.000  |
|advert      |    1.863    |   0.683    |  2.726  |  0.008  |
>
```

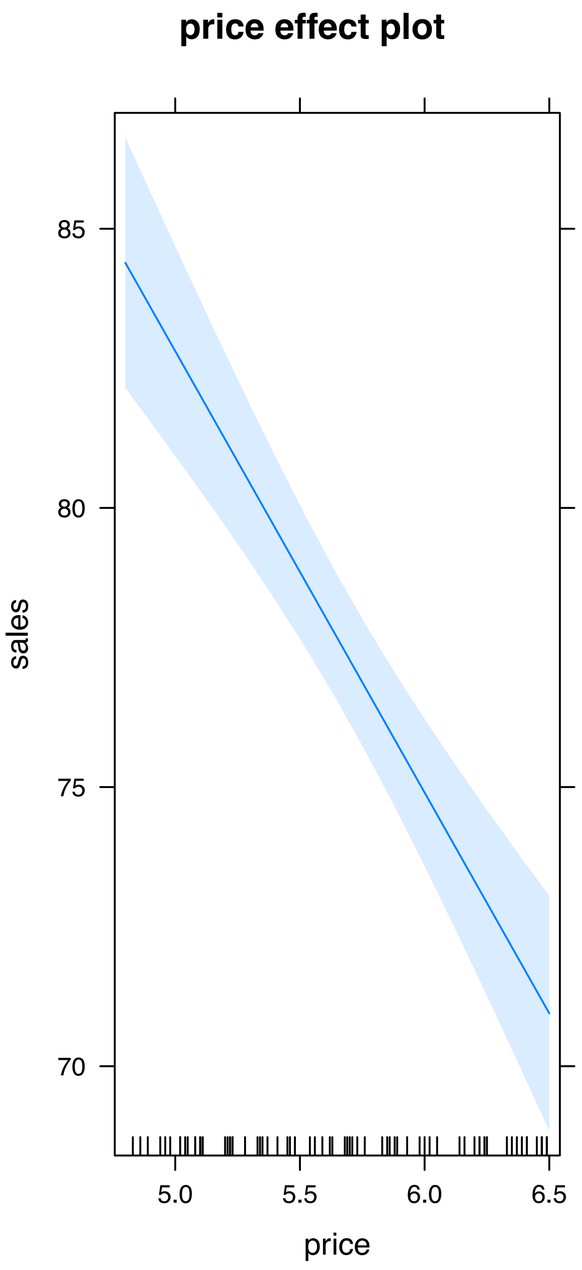
$H_0 : \beta_k = 0$ vs $H_1 : \beta_k \neq 0$

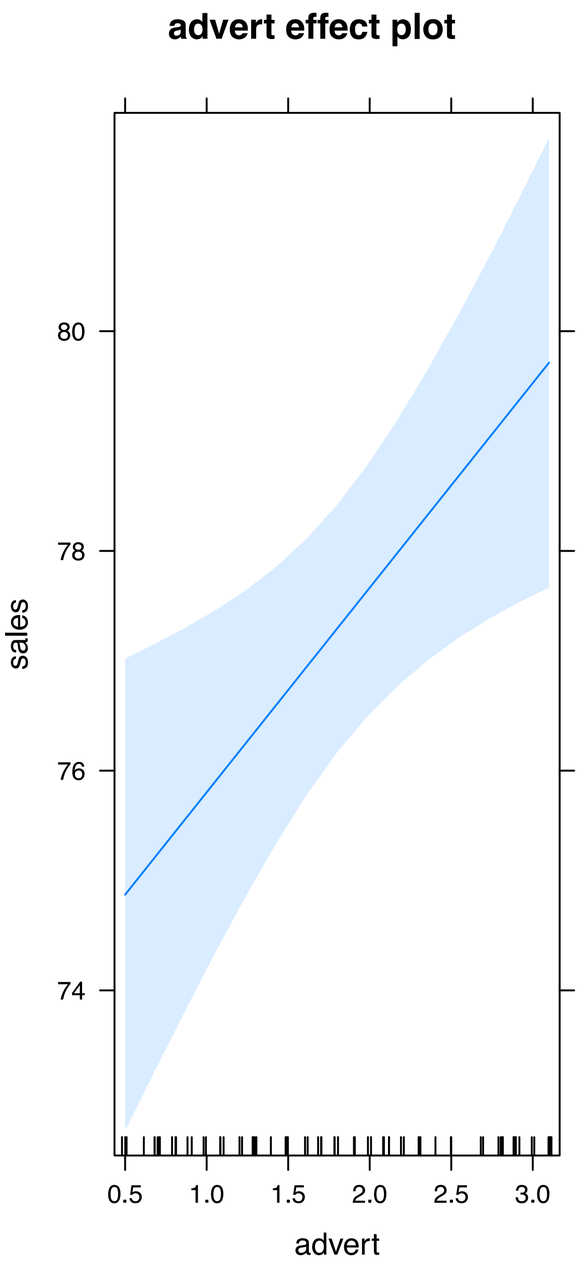
Point estimation: find an estimator
-> Standardize an estimastor
-> CI
-> Hypothesis test

Confidence band

The following left shows the predicted levels of the dependent variable sales and its 95% confidence band for the sample values of the variable price. Function `effect` plots only one graph for one variable, while `allEffects` plots all variables in the model.


price effect plot
85
80
75
70
sales
5.0
5.5
6.0
6.5
price


price effect plot
sales
85
80
75
70
5.0
5.5
6.0
6.5
price

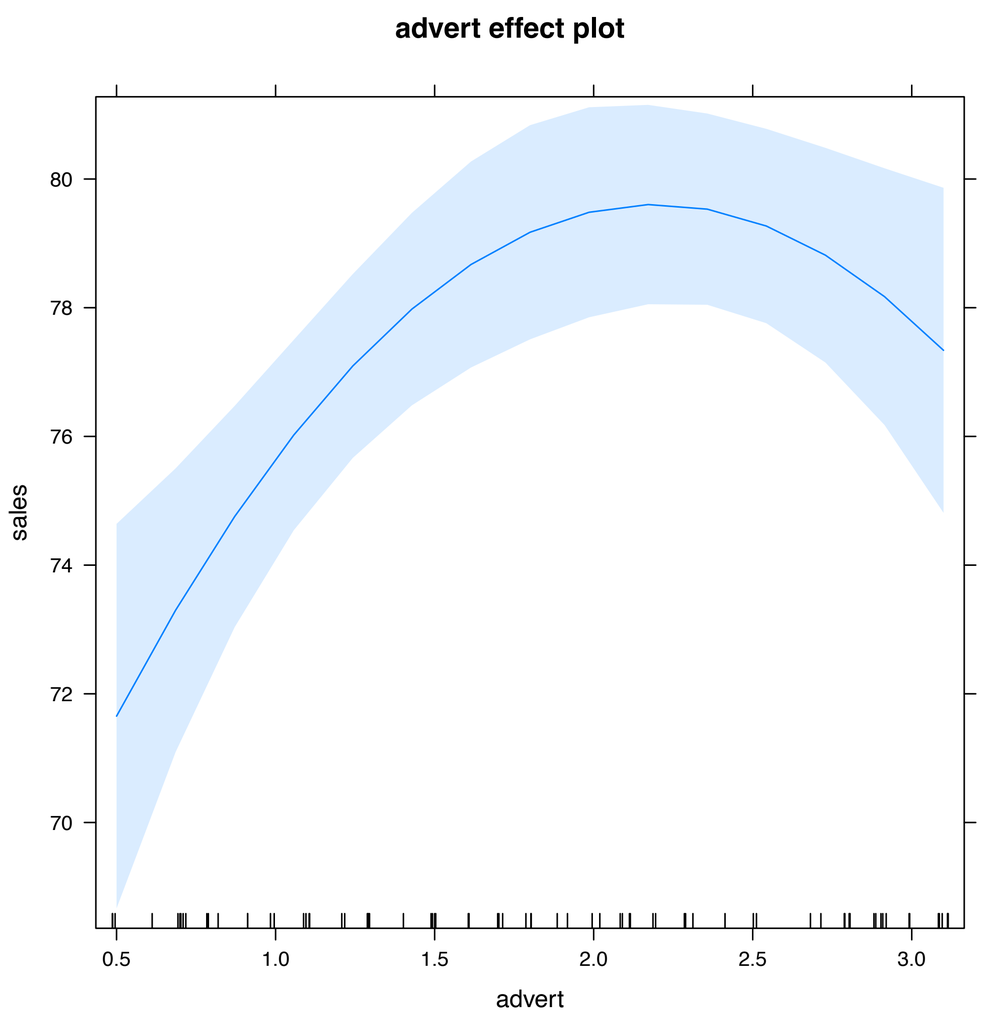

advert effect plot
sales
80
78
76
74
0.5
1.0
1.5
2.0
2.5
3.0
advert

R scripts

```
> effprice <- effect("price", mod1)
Error in pdf("effprice.pdf") : cannot open file 'effprice.pdf'
> plot(effprice)
> summary(effprice)

 price effect
price
     4.8      5.2      5.7      6.1      6.5
84.39052 81.22737 77.27345 74.11030 70.94716

 Lower 95 Percent Confidence Limits
price
     4.8      5.2      5.7      6.1      6.5
82.14947 79.67882 76.14838 72.66864 68.84513

 Upper 95 Percent Confidence Limits
price
     4.8      5.2      5.7      6.1      6.5
86.63156 82.77593 78.39851 75.55197 73.04919
```

```
> alleffandy <- allEffects(mod1)
> plot(alleffandy)
```

Another example of using the function `effect`.



```
> mod2 <- lm(sales~price+advert+I(advert^2), data=andy)
> summary(mod2)

Call:
lm(formula = sales ~ price + advert + I(advert^2), data = andy)

Residuals:
     Min       1Q   Median       3Q      Max 
-12.2553  -3.1430  -0.0117   2.8513  11.8050 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 109.7190     6.7990  16.137  < 2e-16 ***
price        -7.6400     1.0459  -7.304 3.24e-10 ***
advert       12.1512     3.5562   3.417  0.00105 ** 
I(advert^2)  -2.7680     0.9406  -2.943  0.00439 ** 
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

R demo: Langrage multiplier test

```
Residual standard error: 4.645 on 71 degrees of freedom
Multiple R-squared: 0.5082, Adjusted R-squared:  0.4875
F-statistic: 24.46 on 3 and 71 DF, p-value: 5.6e-11

> plot(effect("I(advert^2)", mod2))
```

All the methods available in R for simple linear regression models are available for multiple models as well, such as `coef`, `anova` etc.

5.3 Sampling properties of the least squares estimators

The distribution for b_k

Because LSE are linear functions of dependent variables, they are also normally distributed,

$$b_k \sim N(\beta_k, var(b_k)) \, .$$

Note

$$t = \frac{b_k - \beta_k}{\sqrt{\hat{var}(b_k)}} = \frac{b_k - \beta_k}{se(b_k)} \sim t_{(N-K)} \, .$$

With this information, we can form an interval estimate and a test statistics for hypothesis testing.

The distribution for $\boxed{\sum c_k b_k}$

A plug-in estimator of $\sum c_k \beta_k$

To estimate a linear combination of the coefficients,

$$\lambda = c_1\beta_1 + \cdots + c_K\beta_K,$$

the estimator is

$$\hat{\lambda} = c_1 b_1 + \cdots + c_K b_K\,.$$

The statistic is

$$t = \frac{\hat{\lambda} - \lambda}{se(\hat{\lambda})} = \frac{\sum c_k b_k - \sum c_k \beta_k}{se(\sum c_k b_k)} \sim t_{N-K},$$

where

$$se(\sum c_k b_k) = \sqrt{\hat{var}(\sum c_k b_k)}$$

$$= \sqrt{\sum c_k^2 \hat{\text{var}}(b_k) + \sum_{\substack{i,j \\ i<j}} 2c_i c_j \hat{\text{cov}}(b_i, b_j)}$$

Variance formula

Appendix: LSE in multiple regression and its sampling distribution

Matrix calculation

Transpose

Multiplication

Inverse of an invertible matrix

give me two matrix of size 2 x 2, A and B.

Sure! Here are two 2×2 matrices:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$$

Want to do something fun with them, like multiplication, inversion, or eigenvalue magic? 😄

What is A's transpose?

The transpose of matrix A, denoted as $A^\top$, is obtained by flipping rows into columns.

Given:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

Then the **transpose** is:

$$A^\top = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Just like flipping a pancake—but math-style. 🥞📐 Want to keep going?

Ex:
Z = [1, 1, 1];

Ex: $Z = [1 \quad 1 \quad 1]$
The size of Z is 1×3

$$Z' = Z^\top = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = [1; 1; 1]$$

What is A \times B?

To compute the matrix product $A \times B$, we multiply corresponding rows of A with columns of B:

Given:

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 7 \\ 6 & 8 \end{bmatrix}$$

Let's compute $A \times B$:

$$A \times B = \begin{bmatrix} 1 \cdot 5 + 3 \cdot 6 & 1 \cdot 7 + 3 \cdot 8 \\ 2 \cdot 5 + 4 \cdot 6 & 2 \cdot 7 + 4 \cdot 8 \end{bmatrix} = \begin{bmatrix} 5 + 18 & 7 + 24 \\ 10 + 24 & 14 + 32 \end{bmatrix} = \begin{bmatrix} 23 & 31 \\ 34 & 46 \end{bmatrix}$$

So,

$$A \times B = \begin{bmatrix} 23 & 31 \\ 34 & 46 \end{bmatrix}$$

Like a well-rehearsed dance—step, pivot, multiply! 💃🕺 Want to try $B \times A$ and see if it's the same? (Spoiler: probably not 😏)

Great question! To find the **inverse** of a 2×2 matrix, we use a neat little formula (like a shortcut through the matrix jungle 🌴📐).

Given:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The inverse A^{-1} exists **only if** the determinant $\det(A) = ad - bc \neq 0$.
If it does, then:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Some math!

U is an n-variate normal distribution, denoted by $U \sim N(\mu, \Sigma)$ if it has a joint pdf

$$f(u) = \frac{1}{\sqrt{(2\pi)^n |\Sigma|}} \exp^{-\frac{1}{2}(u-\mu)'\Sigma^{-1}(u-\mu)} .$$

Here, U and μ are $n \times 1$ vectors, Σ is an $n \times n$ covariance matrix.

Consider an affine transformation for U, denoted by $V = C + DU$, where C is an $m \times 1$ vector, D is an $m \times n$ matrix. Then, V is an m-variate normal distribution:

$$V \sim N(C + D\mu, D\Sigma D') .$$

Variance formula in matrix form

Proof: LSE

In vector form, we rewrite the multiple regression model as

$$Y = X\beta + e,$$

where $Y = (y_1, \ldots, y_N)', \beta = (\beta_1, \ldots, \beta_K)', e = (e_1, \ldots, e_N)',$ and

$$X = \begin{pmatrix} 1 & x_{1,2} & \cdots & x_{1,K} \\ 1 & x_{2,2} & \cdots & x_{2,K} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{N,2} & \cdots & x_{N,K} \end{pmatrix},$$

and $e \sim N_N(0,\sigma^2 I)$, where I is the identity matrix.

Note we have an equal expression.

$$Y \sim N(X\beta, \sigma^2 I)\,.$$

We write the sum of squared errors as

$$SSE(\beta) = (Y - X\beta)'(Y - X\beta)\,.$$

The first-order condition requires that

$$\frac{\partial SSE(\beta)}{\partial \beta} = -\,2X'(Y - X\beta) = 0.$$

Thus, $-2X'(Y - X\beta) = 0$ implies that $X'Y = X'X\beta$ and $\beta = (X'X)^{-1}(X'Y)$. Thus, the LSE for β is

$$b = (X'X)^{-1}(X'Y)\,.$$

Recall $Y \sim N(X\beta, \sigma^2 I)$. Because $b = (X'X)^{-1}X'Y$ is an affine transformation for Y and hence a normal distribution.

The expectation:

$$E(b) = E((X'X)^{-1}X'Y) = E((X'X)^{-1}X'(X\beta + e)) = \beta + 0 = \beta,$$

The variance:

$$\begin{aligned}
\mathrm{var}(b) &= \mathrm{var}\left((X'X)^{-1}X'Y\right) \\
&= (X'X)^{-1}X'\mathrm{var}(Y)((X'X)^{-1}X')' \\
&= ((X'X)^{-1}X')(\sigma^2 I)X(X'X)^{-1} \\
&= \sigma^2(X'X)^{-1}\,.
\end{aligned}$$

As a result, we prove

$$b \sim N\left(\beta, \sigma^2(X'X)^{-1}\right),$$

5.4 Interval estimation

- Because of $\dfrac{b_k - \beta_k}{se(b_k)} \sim t_{(N-K)}$

- $1 - \alpha = P\left(-t_{\alpha/2;(N-K)} < \dfrac{b_k - \beta_k}{se(b_k)} < t_{\alpha/2;(N-K)}\right)$

5.4 Interval estimation

Following the same rationale, a $100(1-\alpha)\,\%$ confidence interval for β_k is

$$\left[b_k - t_{(1-\alpha/2,N-K)}se(b_k),\, b_k + t_{(1-\alpha/2,N-K)}se(b_k)\right].$$

Interval estimation is similar, except the number of degrees of freedom is now $(N-K)$.

With the calculated standard error, can now determine a $95\,\%$ confidence interval. The code also shows using the R function `confint(model, parm, level)` to check our results.

R scripts: from scratch

```
> varb1 <- vcov(mod1)[1,1]
> varb2 <- vcov(mod1)[2,2]
> varb3 <- vcov(mod1)[3,3]
> covb1b2 <- vcov(mod1)[1,2]
> covb1b3 <- vcov(mod1)[1,3]
> covb2b3 <- vcov(mod1)[2,3]
> seb2 <- sqrt(varb2) #standard error of b2
> seb3 <- sqrt(varb3)
> alpha <- 0.05
> tcr <- qt(1-alpha/2, df)
> lowb2 <- b2-tcr*seb2
> upb2 <- b2+tcr*seb2
> lowb3 <- b3-tcr*seb3
> upb3 <- b3+tcr*seb3
> cat('b2: [', lowb2, ',', upb2,'],  b3:[', lowb3, '.', upb3);
```

R scripts: confint

```
b2: [ -10.09268 , -5.723032 ],  b3:[ 0.500659 . 3.22451>
> confints <- confint(mod1, parm=c("price", "advert"),
level=0.95)
> kable(data.frame(confints),
+   caption="Confidence intervals for 'price' and 'advert'",
+   align="c", col.names=c("lowb", "upb"), digits=4)


|        |   lowb    |   upb    |
|:------|:---------:|:--------:|
|price  | -10.0927 | -5.7230 |
|advert |  0.5007  | 3.2245  |
>
```

Interested in understanding $\sum c_k \beta_k$.

A plug-in estimator: $\sum c_k b_k$.

$$\sum c_k b_k \sim N(\sum c_k \beta_k, var(c_k b_k))$$

We can standardize it

$$\frac{\sum c_k b_k - \sum c_k \beta_k}{se(\sum c_k b_k)} \sim t_{(N-K)}$$

The $100(1-\alpha)\,\%$ CI for $\sum c_k \beta_k$

$$\sum c_k b_k \pm t_{(\alpha/2;(N-K))} se(\sum c_k b_k)$$

The test statistic for $H_0 : \sum c_k \beta_k = c_0$

$$\frac{\sum c_k b_k - c_0}{se(\sum c_k b_k)} \sim t_{(N-K)}$$

Variance formula of $\sum c_k b_k$

Element-by-element: $\text{var}(\sum c_k b_k) = \sum c_k^2 \text{var}(b_k) + 2 \sum_{i<j} c_i c_j \text{cov}(b_i, b_j)$

To estimate the variace, the plug-in rule:

$$\widehat{\text{var}}\,(\sum c_k b_k) = \sum c_k^2 \hat{\text{var}}(b_k) + 2 \sum_{i<j} c_i c_j \widehat{\text{cov}}\,(b_i, b_j)$$

In matrix form: let $c = (c_1, \ldots, c_K)'$, $b = (b_1, \ldots, b_K)'$. Then, $\sum c_k b_k = c'b$.

The variance formula is $\text{var}(c'b) = c' \times \text{cov}(b) \times c$.

The plug-in rule: $\widehat{\text{var}}\,(c'b) = c' \times \widehat{\text{cov}}\,(b) \times c$

Example: CI of $\sum c_k b_k$

Suppose one is interested in determining an interval estimate for sales when the price decreases by 40 cents and the advertising expenditure increases by \$800 in the andy basic equation:

$$\lambda = 0\beta_1 - 0.4\beta_2 + 0.8\beta_3 \,.$$

Take difference of the following two equations:

$$\text{sales}_1 = \beta_1 + \beta_2 \text{price} + \beta_3 \text{advert},$$
$$\text{sales}_2 = \beta_1 + \beta_2(\text{price} - 0.4) + \beta_3(\text{advert} + 0.8) \,.$$

we obtain $\lambda = \text{sales}_2 - \text{sales}_1 = 0\beta_1 - 0.4\beta_2 + 0.8\beta_3$.

The expected change in sale is approximately 4.653(\$1000s) with 90 % interval estimates [3.47,5.83].

R scripts from scratch

```
## all functions in simple linear regression can be used for multiple regression
mod1 <- lm(sales~price+advert, data=andy)
smod1 <- summary(mod1)
df <- mod1$df.residual
N <- nobs(mod1)
b1 <- coef(mod1)[[1]] #or
b2 <- coef(mod1)[["price"]]
b3 <- coef(mod1)[["advert"]]
```

```
# Show the estimated covariance matrix
vcov(mod1)
kable(data.frame(vcov(mod1)), align='c', digits=3,
      caption="The coefficient covariance matrix",
      col.names=c("(Intercept)", "price", "advert"))

## 5.3 interval estimation
#With the calculated standard error of  b2b2 ,  se(b2)=1.096se(b2)=1.096 , we can now determ
varb1 <- vcov(mod1)[1,1]
varb2 <- vcov(mod1)[2,2]
varb3 <- vcov(mod1)[3,3]
covb1b2 <- vcov(mod1)[1,2]
covb1b3 <- vcov(mod1)[1,3]
covb2b3 <- vcov(mod1)[2,3]
```

R scripts from scratch

```
> alpha <- 0.1
> tcr <- qt(1-alpha/2, df)
> a1 <- 0
> a2 <- -0.4
> a3 <- 0.8
> L <- a1*b1+a2*b2+a3*b3
> varL <- a1^2*varb1+a2^2*varb2+a3^2*varb3+
+   2*a1*a2*covb1b2+2*a1*a3*covb1b3+2*a2*a3*covb2b3
> seL <- sqrt(varL)
> lowbL <- L-tcr*seL
> upbL <- L+tcr*seL
> cat('L: ', L, 'with 90% interval estimate [', lowbL, ', ', upbL, ']')
L:  4.653209 with 90% interval estimate [3.470785 ,  5.835633]
```

R scripts using vector operation

```
# A different way to calculate the variance of L using vector operation.
a <- c(0, -0.4, 0.8) # vector
b <- as.numeric(coef(mod1))# vector of coefficients
L <- sum(a*b) # sum of elementwise products
V <- vcov(mod1) # the variance-covariance matrix
A <- as.vector(a) # (indeed not necessary)
varL <- as.numeric(t(A) %*% V %*% A) # '%*%': matrix multiplication
```

5.5 Hypothesis testing

S.O.P.

The standard procedure for conduct a hypothesis testing:

1. Form a null hypothesis H_0 and an alternative hypothesis H_1.
2. Specify α.
3. Specify a test statistics and its distribution
4. Calculate the sample value of the test statistics.
5. Find the rejection region or the p-value
6. Conclude.

5.5.1. Testing the significance of a single coefficient

When we set up a multiple regression model, we do so because we believe the explanatory variables influence the dependent variable y.

If a given explanatory variable, say x_k, has no bearing on y, then $\beta_k = 0$.

Testing this null hypothesis is sometimes called a **test of significance** for the explanatory variable x_k.

- $H_0 : \beta_k = 0$ vs $H_1 : \beta_1 \neq 0$.
- Set α.
- Under H_0, the test statistic is

$$t_{STAT} = \frac{b_k}{se(b_k)} \sim t_{(N-K)}.$$

5.5.2 One-tail hypothesis testing for a single parameter

Example 5.10 Testing for Elastic Demand

$$E(SALES \mid PRICE, ADVERT) = \beta_1 + \beta_2 PRICE + \beta_3 ADVERT$$

$\beta_2 \geq 0$: a decrease in price leads to a change in sales revenue that is zero or negative (demand is price-inelastic or has an elasticity of unity)

$\beta_2 < 0$: a decrease in price leads to an increase in sales revenue (demand is price-elastic)

1. Set $H_0 : \beta_2 \geq 0$ (demand is unit-elastic or inelastic) vs $H_1 : \beta_2 < 0$ (demand is elastic)
2. Set $\alpha = 0.05$.
3. Under H_0, the test statistic is

$$t_{STAT} = \frac{b_2}{se(b_2)} \sim t_{(N-K)}.$$

4. $t^* = \frac{-7.908}{1.096} = -7.215$.
5. The p-value is $P(t_{72} < -7.215) = 0$. Because p-value $< \alpha$, we reject $H_0 : \beta_2 \geq 0$ and conclude that $H_1 : \beta_2 < 0$ (demand is elastic).

Example: $H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$

```
> alpha <- 0.05
> df <- mod1$df.residual
> tcr <- qt(1-alpha/2, df)
> b2 <- coef(mod1)[["price"]]
> seb2 <- sqrt(vcov(mod1)[2,2])
> t <- b2/seb2
> pval <- 2*(1-pt(abs(t), df)) #two-tail test
> pval
[1] 4.423999e-10
```

5.5.3: testing $\sum c_k \beta_k$

Example 5.12 Testing the Effect of Changes in Price and Advertising

Suppose we wish to determine if lowering the price by 20 cents increases sales more than an increase in advertising expenditure by \$500.

Therefore, we form

$$\begin{aligned} \text{sale}_1 &= \beta_1 + \beta_2(\text{price} - 0.20) + \beta_3 \text{advert}, \\ \text{sale}_2 &= \beta_1 + \beta_2 \text{price} + \beta_3(\text{advert} + 0.5)\,. \end{aligned}$$

Want to know if $H_1 : sale_1 > sale_2$, i.e., $sale_1 - sale_2 > 0$, or,

$$0\beta_1 - 0.2\beta_2 - 0.5\beta_3 > 0.$$

The hypothesis to test this conjecture is

$$H_0 : 0\beta_1 - 0.2\beta_2 - 0.5\beta_3 \leq 0 \;\; vs \;\; H_1 : 0\beta_1 - 0.2\beta_2 - 0.5\beta_3 > 0.$$

If H_0 is true, the test statistic is

$$t_{STAT} = \frac{(-0.2b_2 - 0.5b_3) - 0}{se(-0.2b_2 - 0.5b_3)} \sim t_{72}\,.$$

The realized test statistic is

$$t^* = 1.622.$$

The p-value is $P(t_{72} > 1.622) = 0.0546$.

Because p-value is 0.0546, the data fails to reject the null hypothesis.

Thus, the conjecture that a decrease in price by 20 cents is more effective than an increase in advertising by \$500 is not supported by the data. ★

```
> A <- as.vector(c(0, -0.2, -0.5))
> V <- vcov(mod1)
> L <- as.numeric(t(A) %*% coef(mod1))
> seL <- as.numeric(sqrt(t(A) %*% V %*% A))
> tval <- L/seL
> pval <- 1-pt(tval, df) # the result (p-value)
> pval
[1] 0.05461891
```

R with matrix form

5.6 Nonlinear Relationships

Overview

Sometimes, we are interested in the quadratic

$$y = \beta_1 + \beta_2 x + \beta_3 x^2 + e,$$

or the cubic

$$y = \beta_1 + \beta_2 x + \beta_3 x^2 + \beta_3 x^3 + e\,.$$

For the general polynomial function:

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_p x^p\,.$$

The slope is

$$\frac{dy}{dx} = \alpha_1 + \alpha_2 x + 3\alpha_3 x^2 + \cdots p\alpha_p x^{p-1}\,.$$

Evaluate at a particular value, $x = x_0$, the slope is

$$\frac{dy}{dx}\Big|_{x=x_0} = \alpha_1 + \alpha_2 x_0 + 3\alpha_3 x_0^2 + \cdots p\alpha_p x_0^{p-1}\,.$$

Example: andy

With the added advert quadratic term as:

$$sales_i = \beta_1 + \beta_2 price_i + \beta_3 advert_i + \beta_4 advert_i^2 + e_i \, .$$

Note that the quadratic term is introduced in the model using the function I( ) to indicate the presence of a mathematical operator.

```
> mod2 <- lm(sales~price+advert+I(advert^2),data=andy)
> smod2 <- summary(mod2)
> tabl <- data.frame(xtable(smod2))
> names(tabl) <- c("Estimate",
+                  "Std. Error", "t", "p-Value")
> kable(tabl, digits=3, align='c',
+   caption="The quadratic version of the $andy$ model")
```

	Estimate	Std. Error	t	p-Value
(Intercept)	109.719	6.799	16.137	0.000
price	-7.640	1.046	-7.304	0.000
advert	12.151	3.556	3.417	0.001
I(advert^2)	-2.768	0.941	-2.943	0.004

Marginal effect

Let us calculate the marginal effect of advert on sales for two levels of advert (0.5, 2):

$$\frac{\partial sales}{\partial advert} = \beta_3 + 2\beta_4 advert\,.$$

The calculated marginal effects for the two levels of advertising expenditure are, respectively, 10.77 and 6.62, which shows, as expected, diminishing returns to advertising at any given price.

5.7 Large Sample Properties of the Least Squares Estimator

Non-linear functions of parameters

- Often, marginal effects or other quantities are non-linear functions of the parameters of a regression model.
- Suppose we would like to test a hypothesis

$$H_0 : g(\beta_1, \beta_2) = c \, .$$

The delta method

- Recall the Taylor expansion:

$$g(b_1, b_2) \approx g(\beta_1, \beta_2) + g_1(\beta_1, \beta_2)(b_1 - \beta_1) + g_2(\beta_1, \beta_2)(b_2 - \beta_2)$$

$$\approx g(\boxed{b_1^*}, \boxed{b_2^*}) + g_1(b_1^*, b_2^*)(\boxed{b_1} - \boxed{b_1^*}) + g_2(\boxed{b_1^*}, \boxed{b_2^*})(\boxed{b_2} - \boxed{b_2^*})$$

Non-Random

Random Variable

Random Variable

The *delta method* allows calculating the variance of such quantities using a Taylor series approximation:

$$var(g(b_1, b_2)) \approx g_1^2 var(b_1) + g_2^2 var(b_2) + 2 g_1 g_2 cov(b_1, b_2),$$

where $g_k = \dfrac{\partial g}{\partial \beta_k}$ calculated at the point $\beta_k = b_k^*$

(where b_k^* is the realized statistic calculated from the data)

Let $\hat{g} = g(b_1, b_2)$ be an estimator for $g(\beta_1, \beta_2)$.

The distribution for $g(b_1, b_2)$ is

$$\frac{g(b_1, b_2) - g(\beta_1, \beta_2)}{\sqrt{g_1^2 \hat{\text{var}}(b_1) + g_2^2 \hat{\text{var}}(b_2) + 2 g_1 g_2 \hat{\text{cov}}(b_1, b_2)}} \overset{approx.}{\rightarrow} t_{(N-K)} \, .$$

Or, can be approximately by CLT

$$\frac{g(b_1, b_2) - g(\beta_1, \beta_2)}{\sqrt{g_1^2 \hat{\text{var}}(b_1) + g_2^2 \hat{\text{var}}(b_2) + 2 g_1 g_2 \hat{\text{cov}}(b_1, b_2)}} \overset{d.}{\rightarrow} N(0,1) \, .$$

Example 5.17: The optimal advertising level

Note that we focus on the regression line:

$$sales = \beta_1 + \beta_2 price + \beta_3 advert + \beta_4 advert^2 .$$

To maximize profits, the optimal advertising level satisfies

$$\frac{\partial sales}{\partial advert} = 1.$$

The left-hand side is

$$\frac{\partial sales}{\partial advert} = \beta_3 + 2\beta_4 advert .$$

Therefore, the optimal advertising level, denoted by g, is the solution of $\beta_3 + 2\beta_4 g = 1$, or

$$g := g(\beta_3, \beta_4) = \frac{1 - \beta_3}{2\beta_4} .$$

To use the delta method, we calculate

$$g_3 = \frac{\partial g}{\partial \beta_3} = -\frac{1}{2\beta_4}, \quad g_4 = \frac{\partial g}{\partial \beta_4} = -\frac{1-\beta_3}{2\beta_4^2}.$$

We estimate g, g_3, and g_4, by $\hat{g} = \dfrac{1-b_3}{2b_4}$, $\hat{g}_3 = -\dfrac{1}{2b_4}$, $\hat{g}_4 = -\dfrac{1-b_3}{2b_4^2}$, respectively.

The point estimate of the optimal advertising level is

$$\hat{g} = \frac{1-b_3}{2b_4} = \frac{1-12.1512}{2\times(-2.27696)} = 2.014.$$

With $se(\hat{g}) = 0.1287$, the 95 % interval estimates for $\hat{g}$ is

$$\begin{aligned}
&\left[\hat{g} \pm t_{(0.975,71)}\text{se}(\hat{g}), \hat{g} + t_{(0.975,71)}\text{se}(\hat{g})\right] \\
&= \left[2.014 - 1.994 \times 0.1287, \quad 2.014 + 1.994 \times 0.1287\right] \\
&= \left[1.758, \quad 2.271\right].
\end{aligned}$$

We estimate with 95 % confidence that the optimal level of advertising lies between \$1757 and \$2271.

```
> alpha <- 0.05
> df <- mod2$df.residual
> tcr <- qt(1-alpha/2, df)
> g <- (1-b3)/(2*b4)
> g3 <- -1/(2*b4)
> g4 <- -(1-b3)/(2*b4^2)
> varb3 <- vcov(mod2)[3,3]
> varb4 <- vcov(mod2)[4,4]
> covb3b4 <- vcov(mod2)[3,4]
> varg <- g3^2*varb3+g4^2*varb4+2*g3*g4*covb3b4
> seg <- sqrt(varg)
> lowbg <- g-tcr*seg
> upbg <- g+tcr*seg
> cat('g is estimated as ', g, 'with 95% interval estimates [', lowbg,
', ', upbg, ']');
g is estimated as 2.01434 with 95% interval estimates [1.757673 ,
2.271006]
```

End of Chapter 5